

GUATEMALA LA AURORA, Guatemala

WMO: 786410

Lat: 14.583N

Lon: 90.527W

Elev: 4952

StdP: 12.25

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.6	53.5	42.4	48.0	62.1	44.5	52.1	62.8	31.8	64.8	28.8	63.5	13.1	0	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.4	82.5	64.2	80.8	64.4	79.2	64.3	68.5	76.8	67.7	75.5	67.2	74.7	11.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	116.9	70.7	65.8	114.5	70.2	64.7	110.3	69.2	35.9	77.3	35.2	75.8	34.7	74.7	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	22.4	20.4		47.9	86.6	1.9	1.8	46.5	87.9	45.3	88.9	44.3	89.9	42.8	91.2	
				44.5	71.6	2.2	2.2	42.9	73.1	41.6	74.4	40.4	75.6	38.8	77.2	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	64.9	66.9	68.5	70.9	71.0	69.5	69.3	69.7	69.3	68.2	66.1	65.5
	DBStd	3.01	2.88	2.68	2.58	2.17	2.09	1.85	1.62	1.50	1.82	2.23	2.83	2.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	104	36	12	4	0	0	0	0	0	1	4	22	25
	CDD50	6684	463	472	572	626	652	586	600	610	578	563	482	480
	CDD65	1314	34	64	111	176	188	136	135	145	129	102	54	40
	CDH74	4066	139	285	552	853	685	276	299	352	237	166	95	127
	CDH80	256	3	14	45	105	67	5	3	6	3	2	1	2
Wind	WSAvg	11.0	13.7	12.8	11.9	10.4	8.9	8.4	11.2	9.9	7.6	10.0	13.5	13.4
Precipitation	PrecAvg	44.9	0.2	0.1	0.5	0.9	4.6	9.3	7.0	7.3	9.2	4.3	1.0	0.2
	PrecMax	65.8	0.7	0.5	2.1	3.2	13.1	16.8	14.2	18.0	15.6	9.8	3.4	2.0
	PrecMin	32.0	0.0	0.0	0.0	0.0	1.8	4.2	2.0	1.5	3.9	0.5	0.0	0.0
	PrecStd	8.9	0.2	0.1	0.6	1.0	2.2	3.1	3.1	3.5	3.0	2.3	1.1	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	82.1	83.9	84.8	84.4	80.6	80.4	80.8	80.2	79.1	78.8	79.1
		MCWB	61.7	61.6	63.1	63.4	65.5	66.7	65.6	64.9	66.4	65.9	64.7	62.6
	2%	DB	77.2	79.1	80.9	82.6	82.1	78.8	78.7	78.9	78.6	77.3	76.8	77.1
		MCWB	61.6	62.2	63.0	64.2	65.2	66.0	64.7	64.9	66.0	65.5	64.0	62.7
	5%	DB	75.3	77.2	79.0	80.8	80.3	77.1	77.1	77.3	76.9	75.6	75.0	75.2
		MCWB	61.4	61.8	62.7	64.4	65.4	65.6	64.7	65.0	65.9	65.0	63.3	62.1
	10%	DB	73.4	75.3	77.2	79.0	78.4	75.4	75.5	76.1	75.1	73.8	73.2	73.3
		MCWB	60.5	61.3	62.3	63.9	65.3	65.3	64.4	64.8	65.4	64.5	62.8	61.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.8	65.9	67.5	69.1	69.6	69.1	68.0	68.2	68.7	68.3	67.1	66.2
		MCDB	74.8	76.3	77.7	79.2	78.6	76.8	75.9	75.5	76.3	75.8	74.1	74.4
	2%	WB	63.4	64.6	66.0	67.5	68.4	67.8	66.9	67.2	67.5	67.1	65.6	64.5
		MCDB	73.0	74.4	75.1	76.9	76.8	75.0	74.5	74.3	74.6	74.0	72.2	73.2
	5%	WB	62.3	63.5	64.9	66.6	67.4	67.0	66.1	66.4	66.8	66.3	64.6	63.2
		MCDB	71.4	72.8	73.9	75.2	75.2	73.8	73.3	73.3	73.6	72.7	71.5	71.5
	10%	WB	61.5	62.3	63.9	65.6	66.7	66.2	65.3	65.6	66.0	65.5	63.8	62.2
		MCDB	70.5	71.0	72.8	73.7	73.7	72.5	72.3	72.3	72.3	71.4	70.3	70.0
Mean Daily Temperature Range	5% DB	MDBR	17.9	19.1	19.9	19.1	16.4	13.7	14.3	14.7	13.6	13.7	14.9	16.8
		MCDBR	20.4	21.2	21.7	20.6	18.7	15.4	15.7	16.1	15.6	15.2	16.9	19.3
		MCWBR	7.9	7.8	7.9	7.0	6.8	5.9	5.6	5.9	6.0	6.2	7.1	7.7
	5% WB	MCDBR	18.8	20.1	19.9	19.0	16.9	14.4	14.3	14.7	14.3	13.8	14.8	17.6
		MCWBR	7.9	8.1	8.0	7.3	6.9	6.2	6.0	6.0	6.1	6.0	6.7	7.8
Clear-Sky Solar Irradiance	taub	0.313	0.328	0.363	0.433	0.462	0.393	0.390	0.380	0.380	0.355	0.331	0.313	
	taud	2.603	2.552	2.424	2.214	2.169	2.419	2.427	2.445	2.429	2.543	2.577	2.611	
	Ebn at Noon	314	312	301	279	266	283	284	290	292	299	305	311	
	Edh at Noon	30	33	38	47	48	37	37	37	37	33	30	29	
All-Sky Solar Radiation	RadAvg	1627	1783	1900	1850	1621	1610	1807	1774	1583	1495	1570	1573	
	RadStd	59	85	93	103	104	94	116	117	114	117	75	72	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page